COMPUTER NETWORKS - UE24CS252B

Lab 5: Understanding Ethernet and ARP using Wireshark

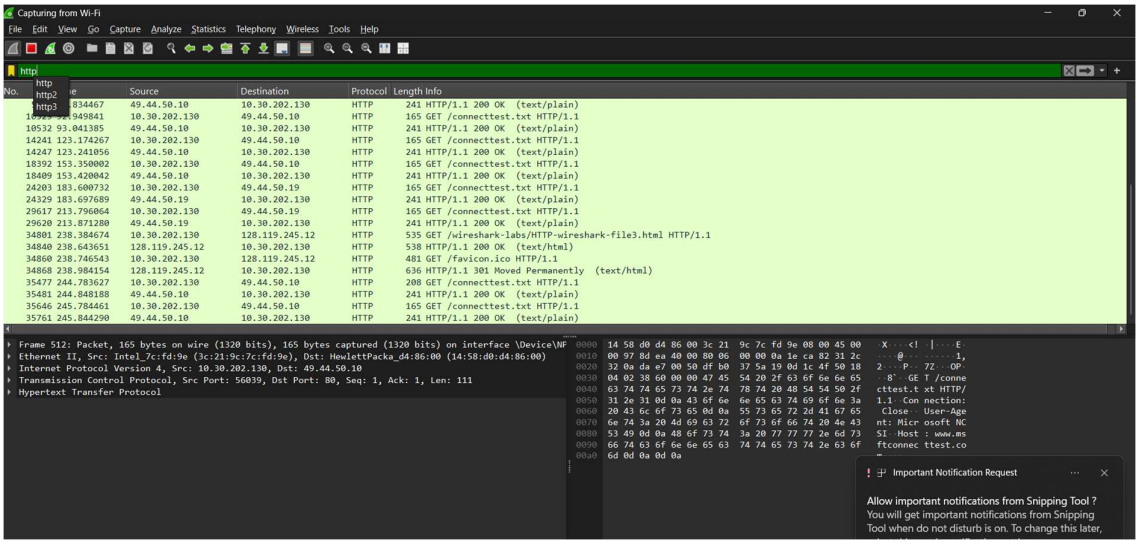
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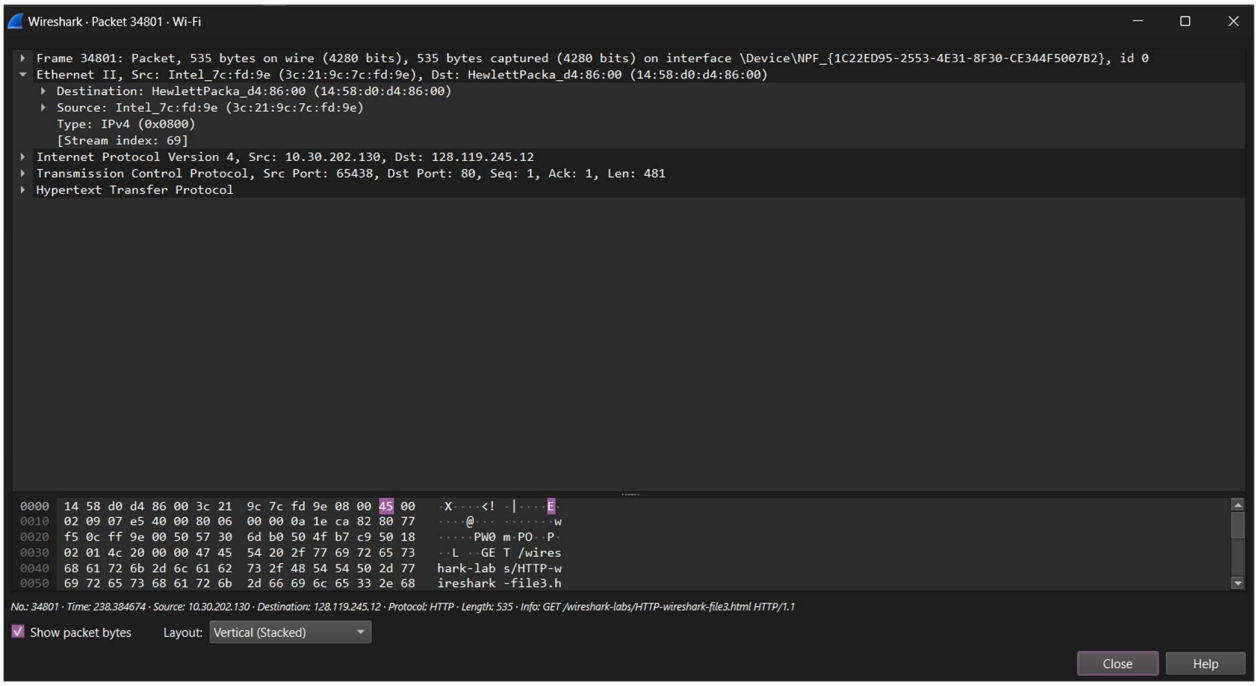
NAME: Tanmayi Nagabhairava

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Part A: Understanding Ethernet

1. HTTP GET Screenshot





2. Ethernet Source Address

- **Source Address:** `3c:21:9c:7c:fd:9e` (Intel_7c:fd:9e).
- **Physical Address:** Yes, this is the MAC address of your device's network interface.

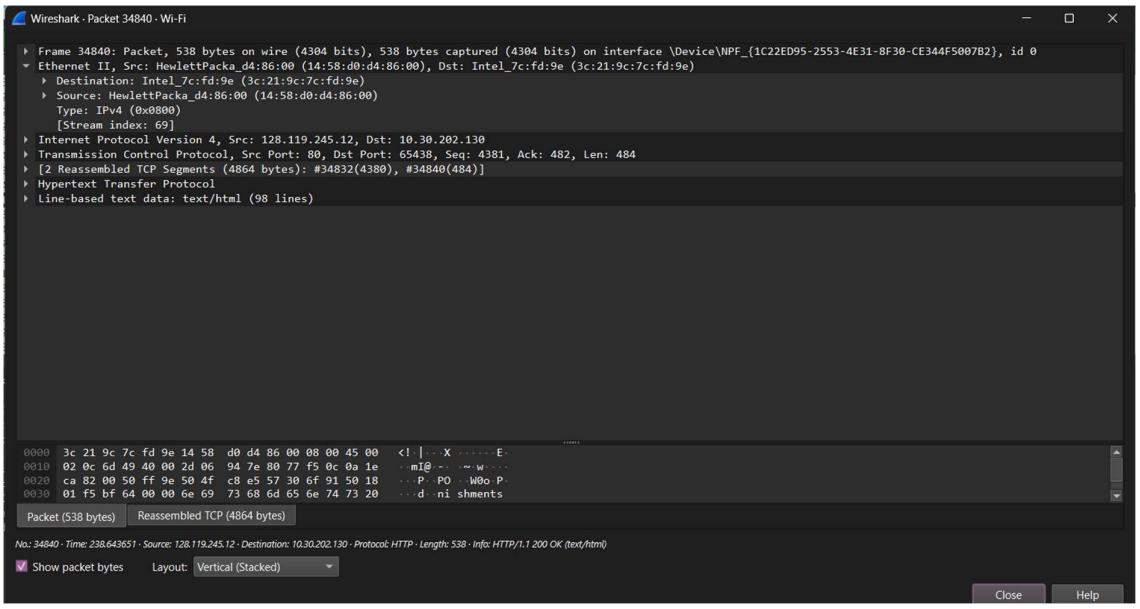
3. Ethernet Destination Address

- **Destination Address:** `14:58:d0:d4:86:00` (HewlettPacka_d4:86:00).
- **Physical Address of Server?** No.
- **Justification:** This MAC address belongs to your **default gateway (router)**. Ethernet frames only travel between devices on the same local segment; your router receives the frame and forwards the IP packet toward the UMass server.

4. Destination IP Address

- **Destination IP:** `128.119.245.12`.
- **Corresponds to Server?** Yes, this is the IP address for the server `gaia.cs.umass.edu`.

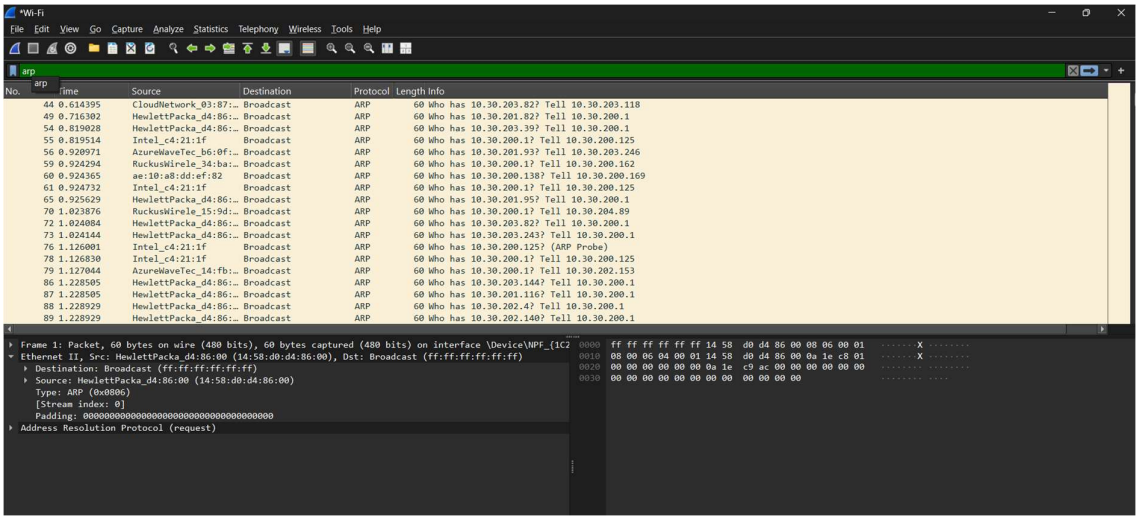
5. HTTP Response Screenshot



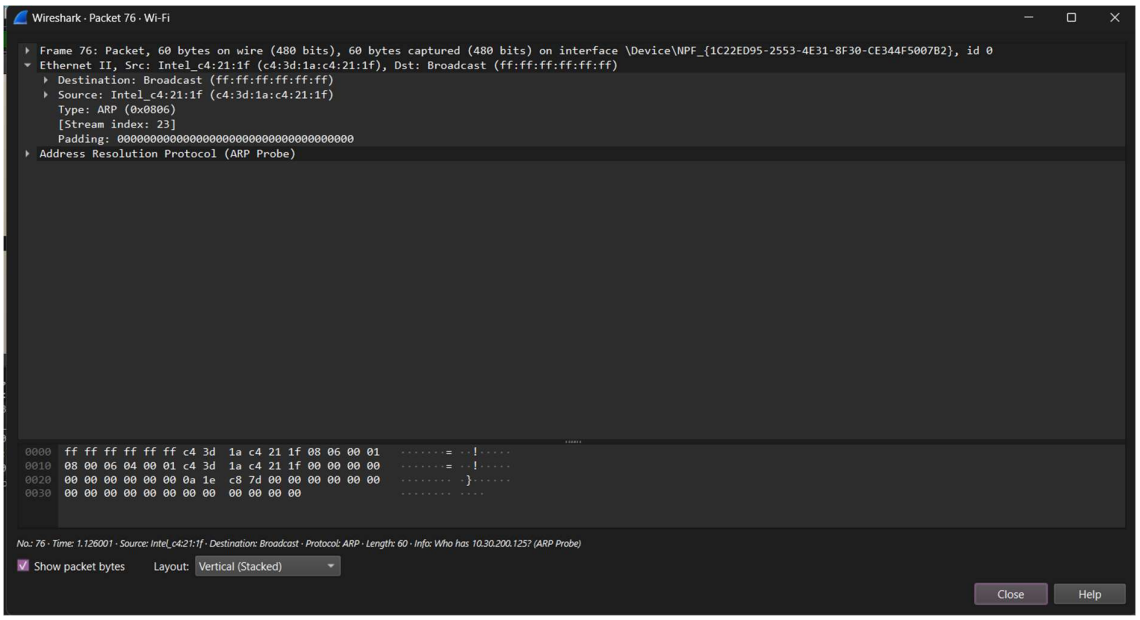
6. Comparison and Changes

- **Address Swap:** The Ethernet source and destination addresses are **reversed** compared to the GET packet.
- **Source Address:** Now `14:58:d0:d4:86:00`, which belongs to your **router**.
- **Destination Address:** Now `3c:21:9c:7c:fd:9e`, which is your **own device**.

Part B: Understanding ARP

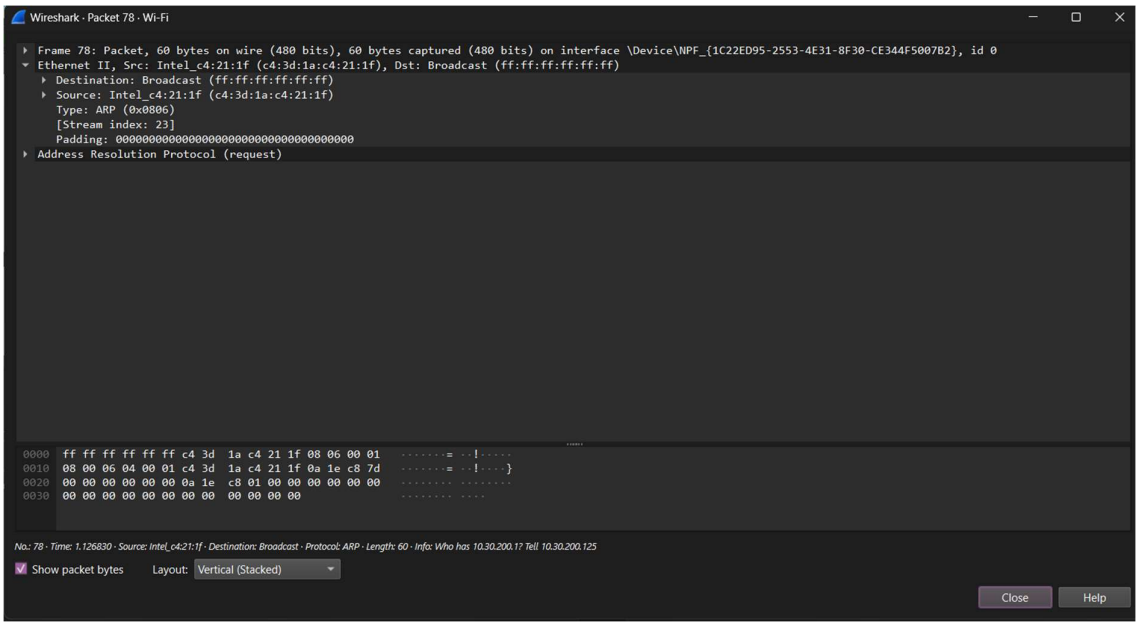


1.ARP Probe Packet



- **Sender IP:** `0.0.0.0`.
- **Target IP:** `10.30.200.125`.
- **Why it's transmitted:** It is used for **IPv4 Address Conflict Detection**. Your device sends this to ensure no other device on the network is already using the IP address it intends to take.

2: ARP Request Packet



- **A. What differences do you observe when compared with the ARP probe packet?**

    - In the **ARP Probe** (Packet 76), the **Sender IP address** is set to `0.0.0.0` because the device hasn't officially claimed the IP yet.
    - In the **ARP Request** (Packet 78), the **Sender IP address** is now set to your device's actual IP: `10.30.200.125`.
- **B. What is the target MAC address in this packet? Justify your answer.**
    - **Target MAC Address:** `00:00:00:00:00:00` (00:00:00:00:00:00).
    - **Justification:** The device does not yet know the physical MAC address of the target (`10.30.200.1`); the primary purpose of an ARP request is to broadcast a query to find this unknown address.
- **C. Which opcode value in the packet indicates that it is an ARP request?**
    - The **Opcode** value is **1**.
    - (In Wireshark, this is listed under the Address Resolution Protocol section as `request (1)`).

3: ARP Response Packet

- **A. What changes do you observe compared to the ARP request packet?**
    - The **Sender** and **Target** fields have been swapped: the device that was the "Target" in the request (the gateway) is now the "Sender" of this reply.
    - The **Sender IP** is now `10.30.200.1` and the **Target IP** is your device `10.30.200.125`.
    - The **Info** column changes from a query ("Who has...?") to a functional statement: `10.30.200.1 is at 14:58:d0:d4:86:00`.
- **B. What is the target MAC address in this packet? Justify your answer.**
    - **Target MAC Address:** `3c:21:9c:7c:fd:9e` (Your device's MAC).
    - **Justification:** ARP replies are **unicast** messages. Unlike the request, which is a broadcast sent to every device (`ff:ff:ff:ff:ff:ff`), the reply is sent directly back to the specific MAC address that requested the information to reduce network congestion.
- **C. Which opcode value indicates that the packet is an ARP reply?**
    - The **Opcode** value is **2**.
    - (In Wireshark, this is listed under the Address Resolution Protocol hardware details as `reply (2)`).